Attention Is All You Need

- NeurIPS 17 -

Kookmin Univ.
Se Won Hong

V. Vaswani et al., "Attention Is All You Need," in Proc. 31st Int. Conf. Neural Inf. Process. Syst. (NeurIPS), Long Beach, CA, USA, 2017,

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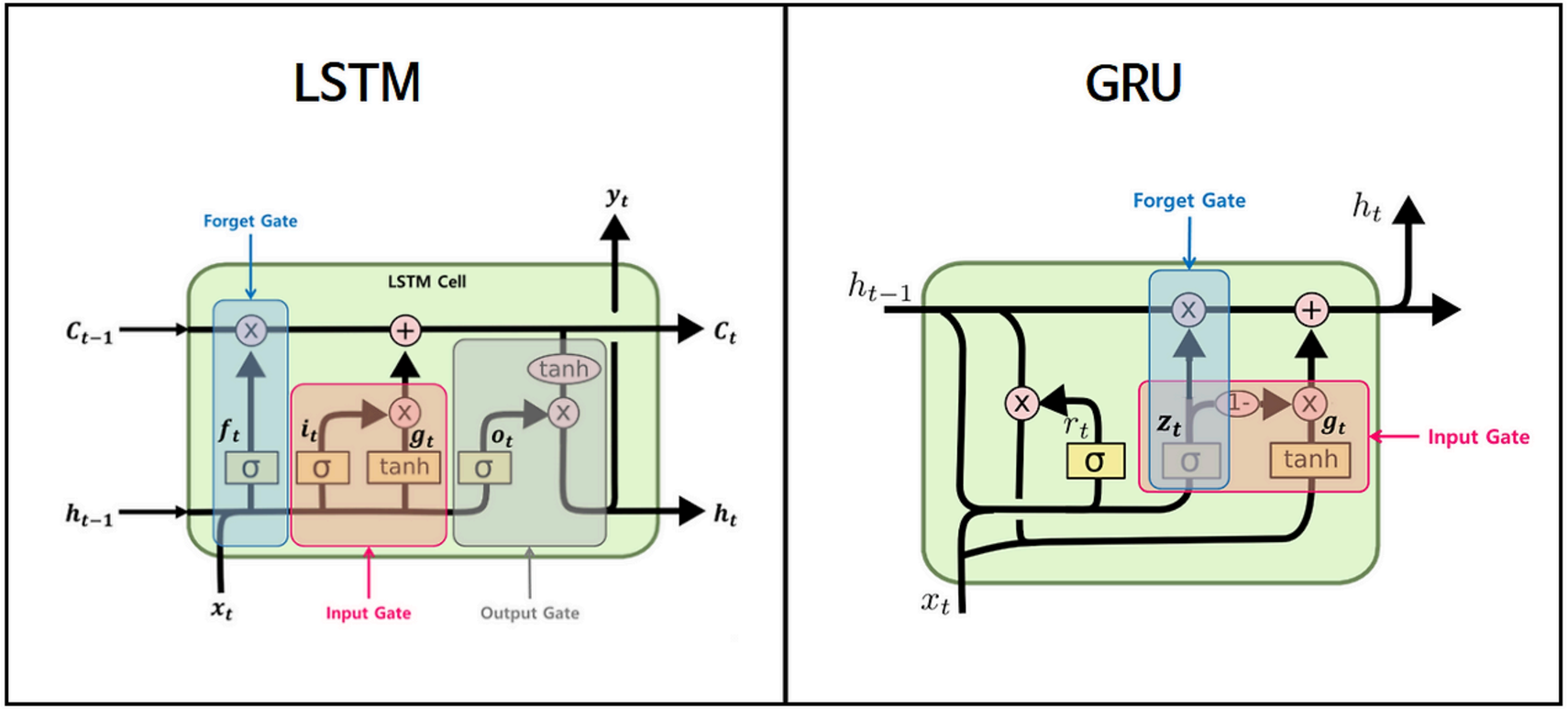
Introduction

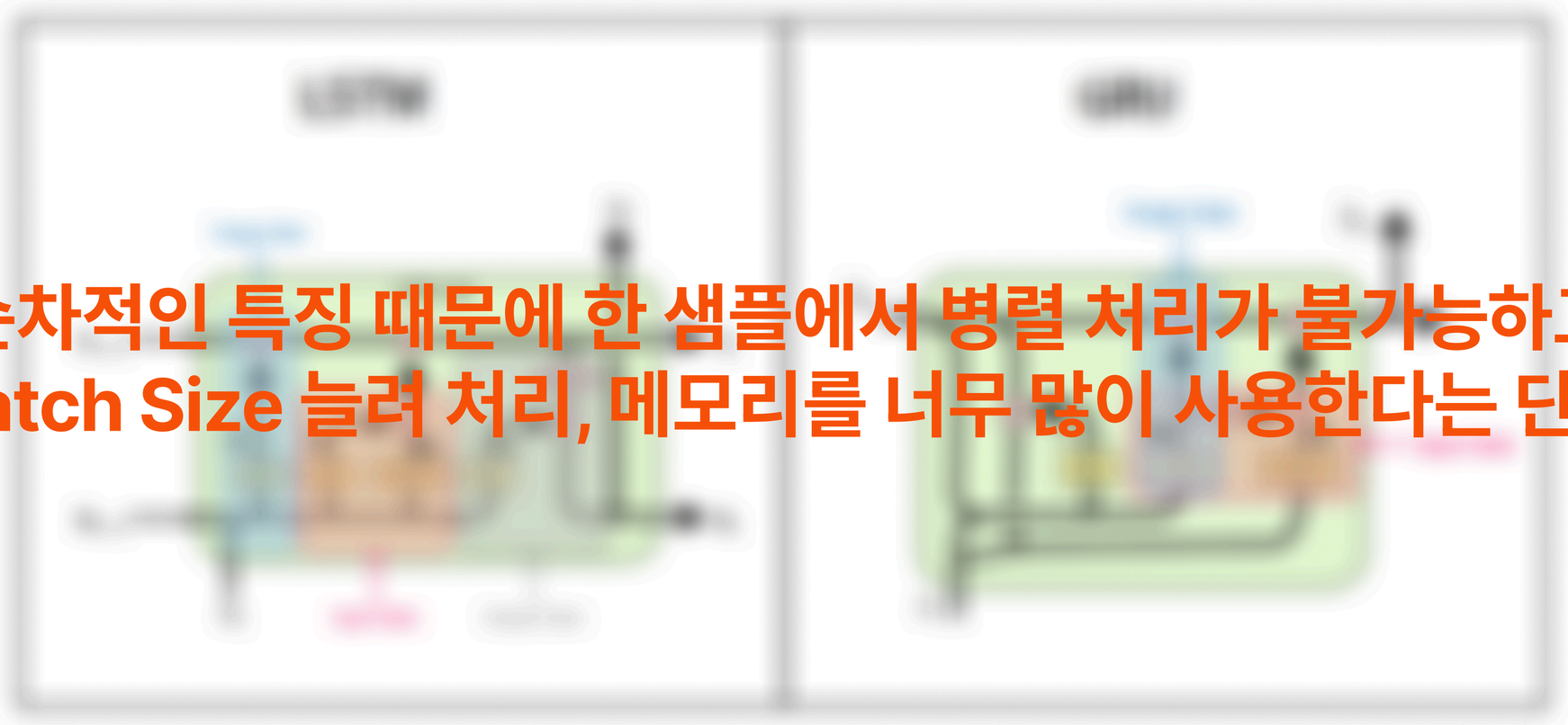
Introduction

이전 연구(Recurrent Models)



Introduction

이전 연구(Recurrent Models)

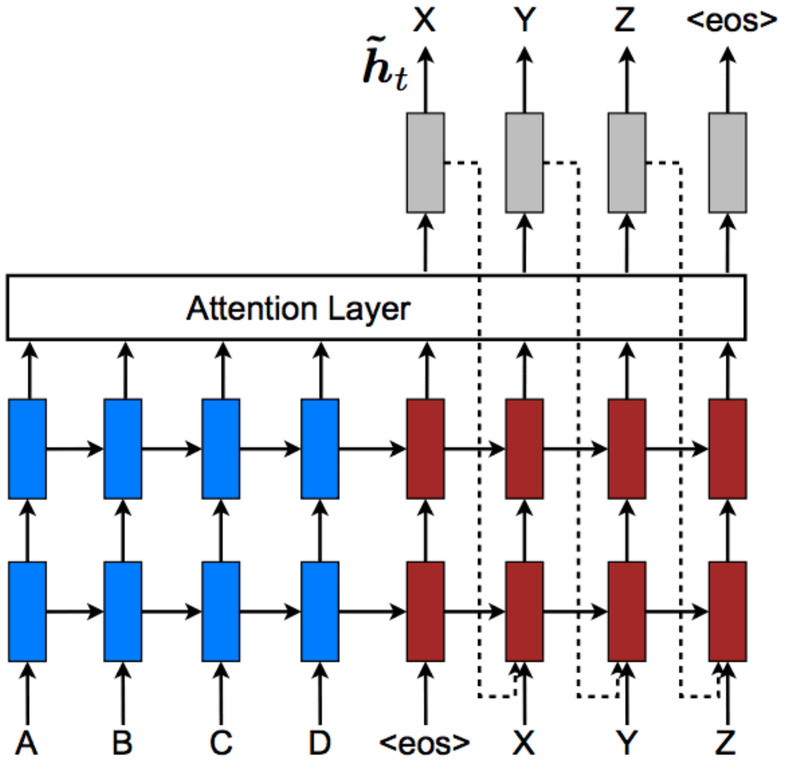


순차적인 특징 때문에 한 샘플에서 병렬 처리가 불가능하고
Batch Size 늘려 처리, 메모리를 너무 많이 사용한다는 단점

Introduction

이전 연구(Recurrent Models) + Attention mechanism

- 성능 향상을 위한 보조 구성 요소로서의 어텐션 메커니즘
- 순차 연산에 기반한 병렬 처리의 한계

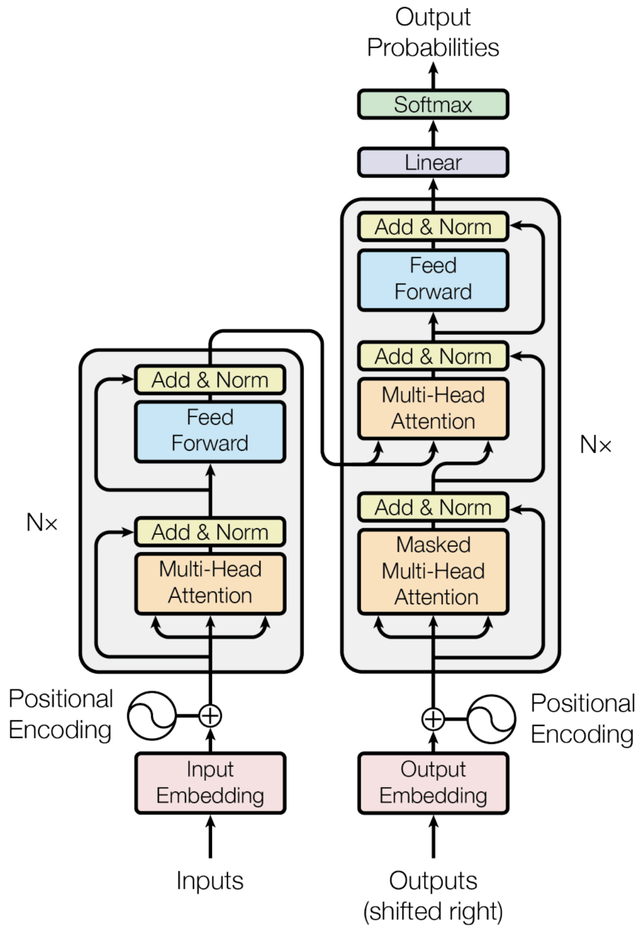


Model

Model

Transformer

- 트랜스포머 : 인코더 구조 + 디코더 구조
- 인코더 : 6개 레이어
- 디코더 : 6개 레이어



Model

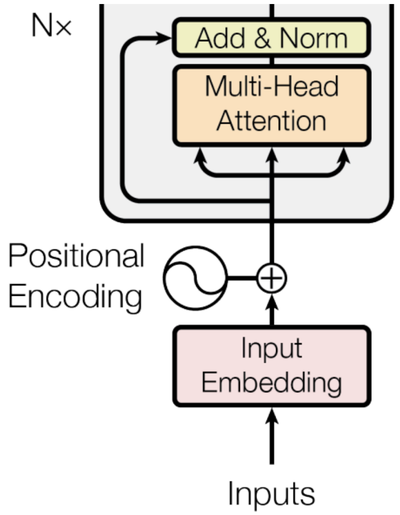
Encoder

First Layer : Multi-Head Self-Attention

- **Self connection, LayerNorm**
- **LayerNorm(x + Sublayer(x))**

Second Layer : position-wise FFNN

- **Self connection, LayerNorm**
- **LayerNorm(x + Sublayer(x))**



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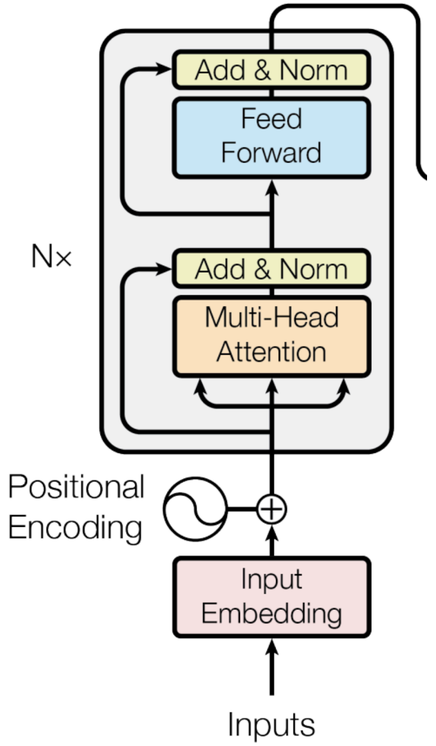

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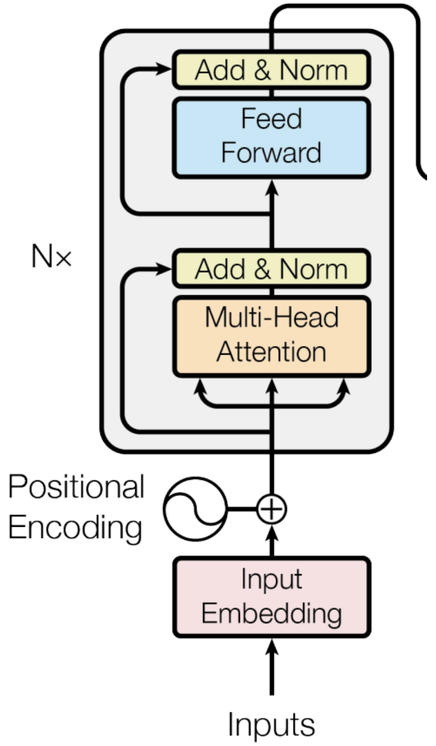
X N



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Second Layer : Encoder - Decoder Cross Attention

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Thired Layer : position-wise FFNN

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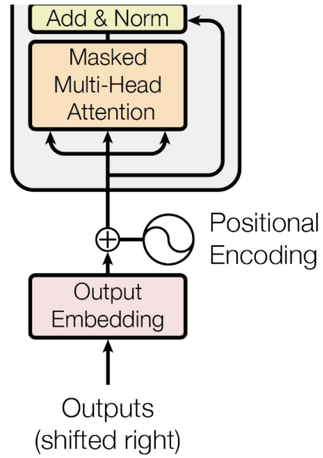

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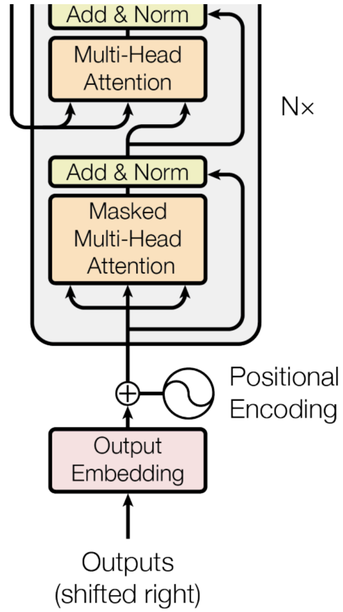

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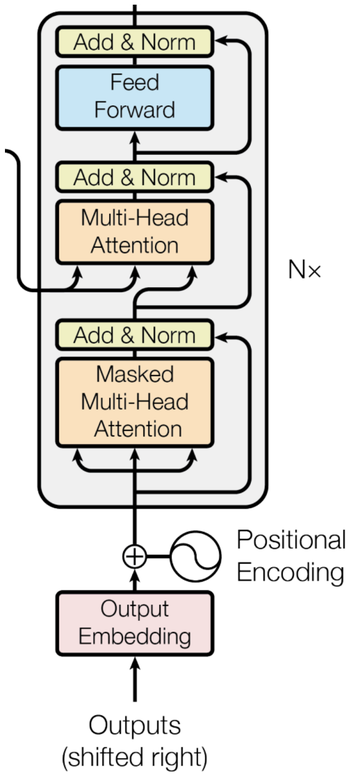

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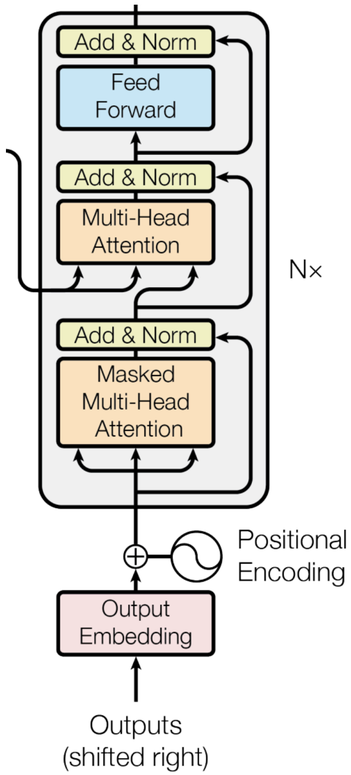


Model

Scaled Dot-Product Attention

X : n x d

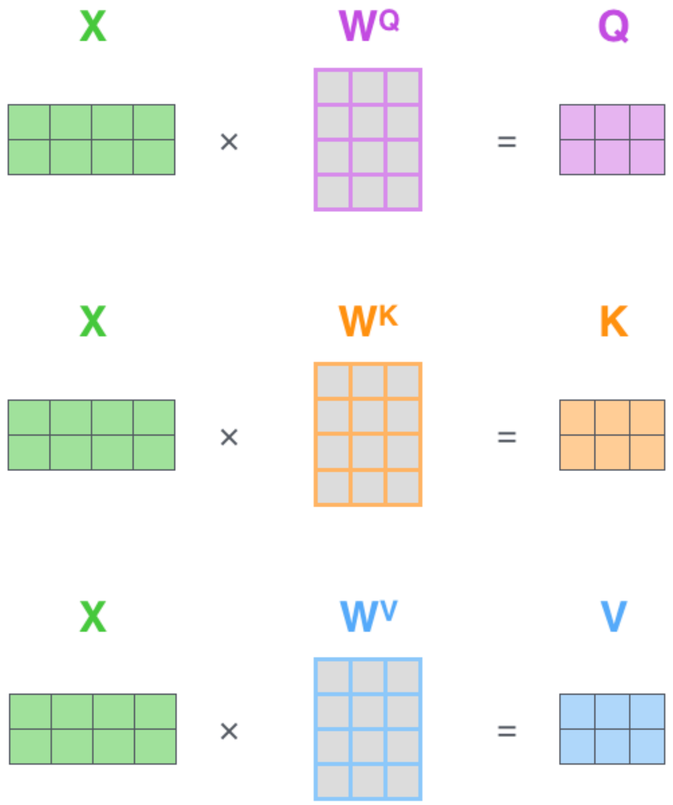
W_q,k : d x d_k

W_v : d x d_v

Q = X x W_q

K = X x W_k

V = X x W_v



Model

Scaled Dot-Product Attention

Q = n x d_k

K = n x d_k

V = n x d_v

$$\text{Attention}(Q, K, V) = \text{softmax}(\frac{QK^T}{\sqrt{d_k}})V$$

Model

Scaled Dot-Product Attention

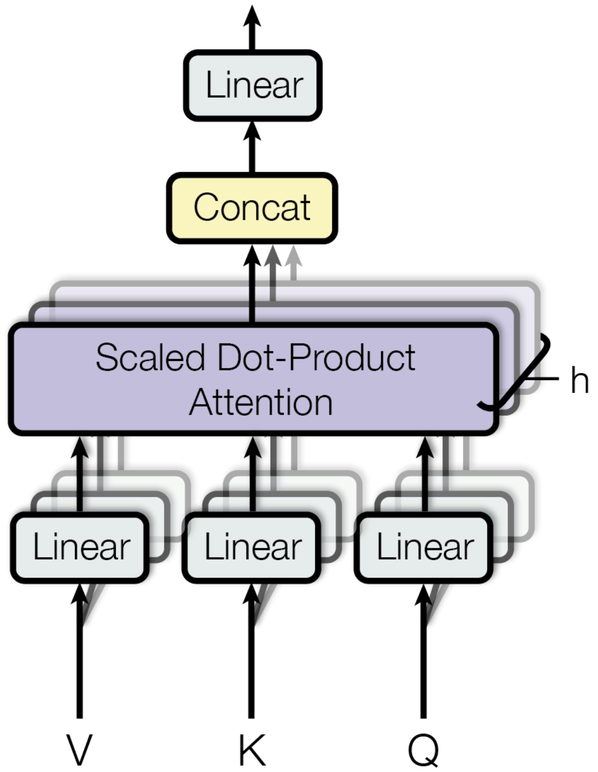
- d_k값이 커질 수록 QK^T값이 매우 커져 기울기 값이 0으로 수렴하게 됨
- 이러한 현상을 방지하기 위해 √d_k를 통해 크기 조절

$$\text{Attention}(Q, K, V) = \text{softmax}(\frac{QK^T}{\sqrt{d_k}})V$$

Model

Multi-head Attention

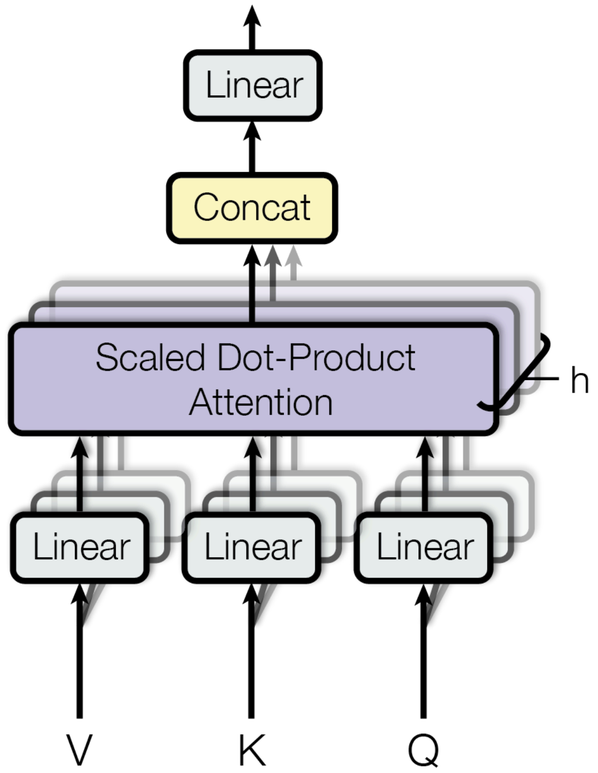
- 입력 시퀀스로부터 Q, K, V 행렬 생성
  - (seq_len × d_model)
- Q, K, V를 각각 h개의 head로 분할 및 선형 투영
  - Q, K: seq_len × {(d_k)/h}, V: seq_len × {(d_v)/h}
- 각 head에서 어텐션 연산 수행
  - (seq_len × (d_v)/h) × h개
- h개의 어텐션 출력을 concat
  - seq_len × d_v
- concat된 출력에 W_o를 곱해 최종 출력 생성
  - 선형 변환을 통한 출력 조정 (크기: (seq_len × d_model))



Model

Multi-head Attention

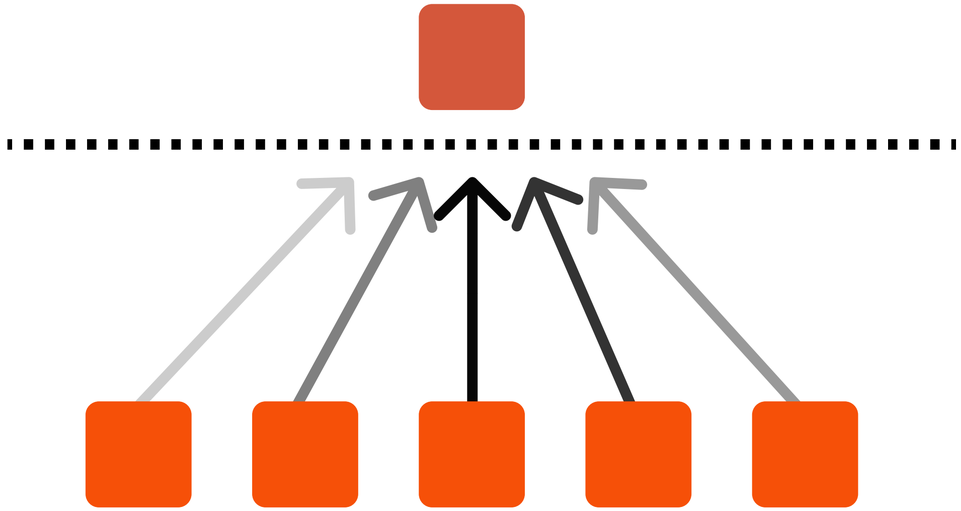
$$\text{MultiHead}(Q, K, V) = \text{Concat}(\text{head}_1, ..., \text{head}_\text{h})W^O$$
$$\text{where head}_\text{i} = \text{Attention}(QW_i^Q, KW_i^K, VW_i^V)$$



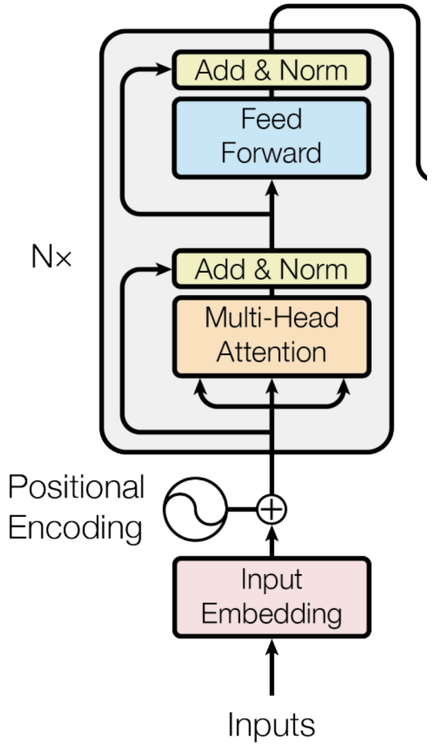
Model

Multi-head Attention - Encoder

Encoder에서의 Q, K, V는 이전 레이어 출력으로부터 생성된 벡터



Self-Attention

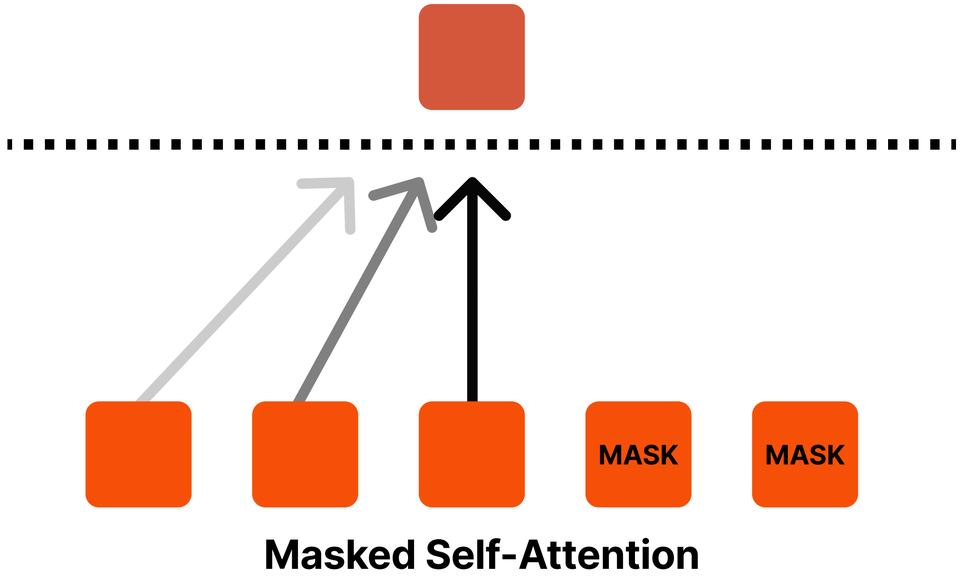


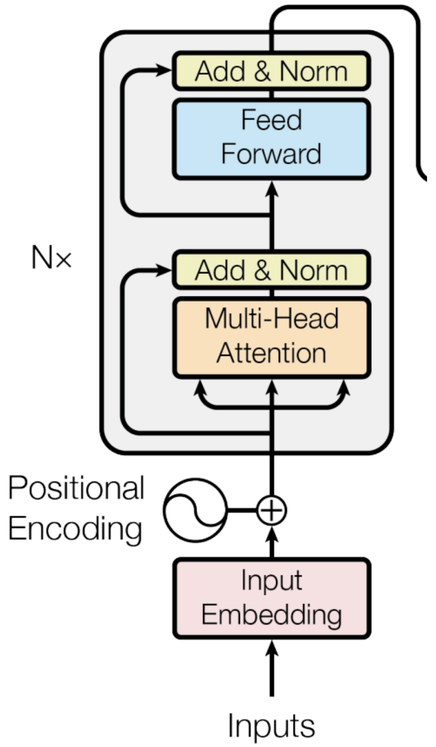
Model

Multi-head Attention - Decoder

Decoder에서의 Q, K, V는 이전 레이어 출력으로부터 생성된 벡터

Decoder는 미래 정보를 MASK하여 정보 흐름을 막음

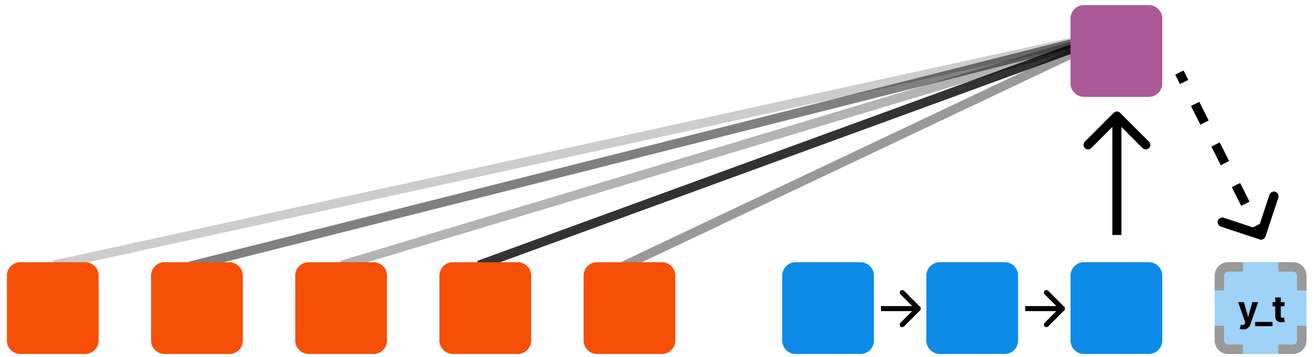




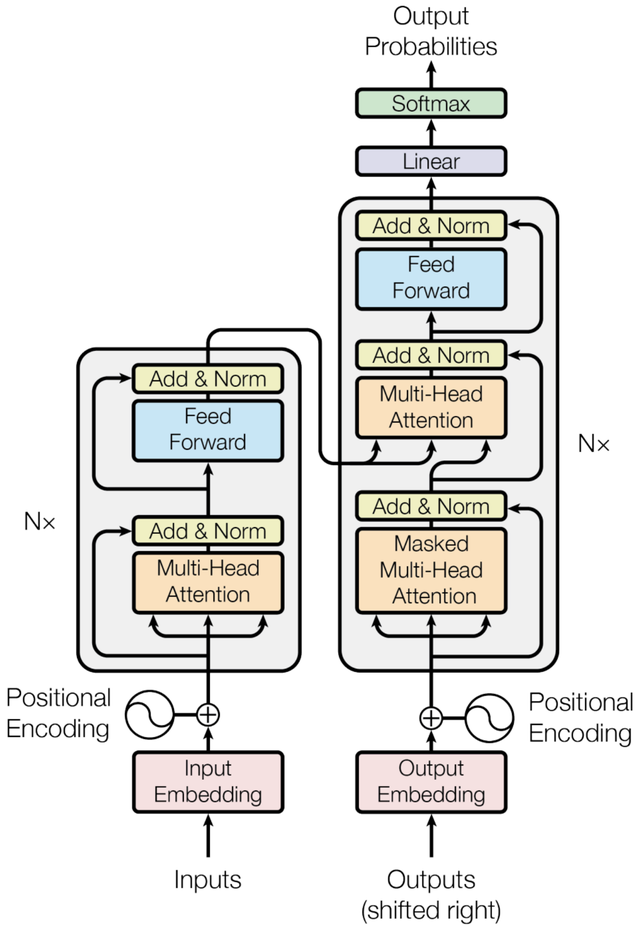
Model

Multi-head Attention - Encoder / Decoder

Encoder/Decoder 경우 Q는 Decoder에서,
K와 V는 Encoder 이전 레이어 출력으로부터 생성된 벡터



Cross-Attention



Model

Position-wise Feed-Forward Networks

- Encoder와 Decoder 각 레이어는 위치별 동일하게 적용되는 FFNN 포함
- 두 개의 선형 변환과 ReLU로 구성된 비선형 변환 구조
- 레이어별로 다른 파라미터를 사용하는 d_model 512, d_ff 2048 구조

$$\mathrm{FFN}(x) = \max(0, xW_1 + b_1)W_2 + b_2$$

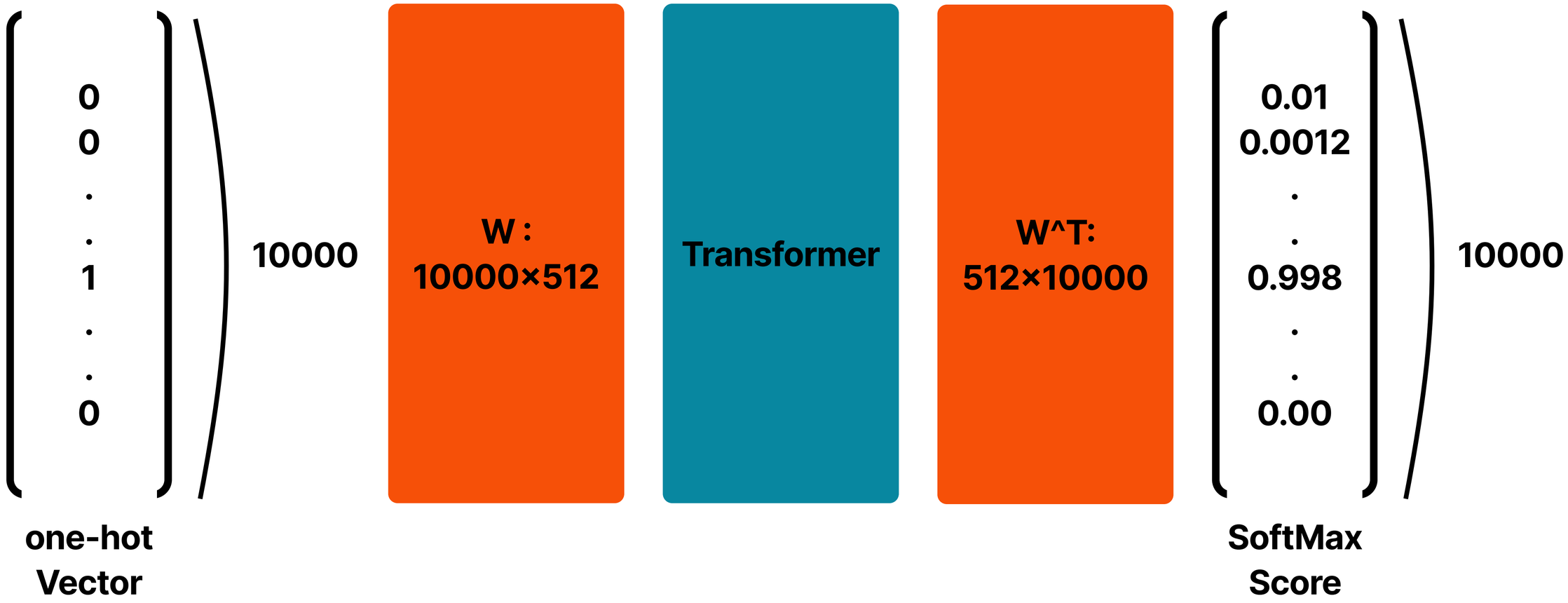
Model

Embeddings and Softmax

- 입력과 출력 토큰을 d_model 차원의 벡터로 변환하기 위해 학습된 W을 사용
- Decoder 출력은 선형 변환과 소프트맥스를 통해 다음 토큰의 확률 분포로 변환됨
- Decoder 출력의 선형 변환에 사용되는 W는 입력 W행렬과 가중치를 공유함
- 스케일 조정을 위해 입력 임베딩 벡터에 √d_model를 곱함

Model

Embeddings and Softmax



Model

Positional Encoding

- Transformer에 들어가는 Sequence에는 위치 정보가 반영되어 있지 않음
- 이를 반영하기 위해 Positional Encoding을 사용함

- pos : sequence 위치
- i : 차원 인덱스
- d_model : sequence 임베딩 차원

$$PE_{(pos,2i)} = sin(pos/10000^{2i/d_{\text{model}}})$$
$$PE_{(pos,2i+1)} = cos(pos/10000^{2i/d_{\text{model}}})$$

Model

Why Self-Attention

Layer Type	Complexity per Layer	Sequential Operations	Maximum Path Length
Self-Attention	$O(n^2 \cdot d)$	$O(1)$	$O(1)$
Recurrent	$O(n \cdot d^2)$	$O(n)$	$O(n)$
Convolutional	$O(k \cdot n \cdot d^2)$	$O(1)$	$O(log_k(n))$
Self-Attention (restricted)	$O(r \cdot n \cdot d)$	$O(1)$	$O(n/r)$

n : 시퀀스 길이
d : 임베딩 사이즈
k : 커널 사이즈
r : 주변 이웃 사이즈

Experiment

Experiment

Model	BLEU		Training Cost (FLOPs)	
	EN-DE	EN-FR	EN-DE	EN-FR
ByteNet [18]	23.75			
Deep-Att + PosUnk [39]		39.2		$1.0 \cdot 10^{20}$
GNMT + RL [38]	24.6	39.92	$2.3 \cdot 10^{19}$	$1.4 \cdot 10^{20}$
ConvS2S [9]	25.16	40.46	$9.6 \cdot 10^{18}$	$1.5 \cdot 10^{20}$
MoE [32]	26.03	40.56	$2.0 \cdot 10^{19}$	$1.2 \cdot 10^{20}$
Deep-Att + PosUnk Ensemble [39]		40.4		$8.0 \cdot 10^{20}$
GNMT + RL Ensemble [38]	26.30	41.16	$1.8 \cdot 10^{20}$	$1.1 \cdot 10^{21}$
ConvS2S Ensemble [9]	26.36	**41.29**	$7.7 \cdot 10^{19}$	$1.2 \cdot 10^{21}$
Transformer (base model)	27.3	38.1	$\mathbf{3.3 \cdot 10^{18}}$	
Transformer (big)	**28.4**	**41.8**	$2.3 \cdot 10^{19}$	